

GLP-1 Receptor Agonist Use Is Associated with Small Intestinal Bacterial Overgrowth and Intestinal Methanogen Overgrowth

Foregut

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DOI: 10.1177/26345161251353437

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Abstract

Introduction: Glucagon-like peptide-1 (GLP-1) receptor agonists (GLP1RAs) are increasingly used to treat obesity and other metabolic conditions; however, they are known to inhibit gastrointestinal motility. Since gastrointestinal microbial overgrowth (MO) can arise in states of intestinal stasis, we sought to investigate the potential association between GLP1RA use and MO.

Methods: We analyzed data from 99 patients prescribed GLP1RAs who subsequently underwent testing for MO via intestinal aspirate culture or glucose or lactulose breath tests.

Results: Most patients had diabetes. Among patients who underwent culture, 76.2% tested positive for small intestinal bacterial overgrowth (SIBO) at a threshold of $\geq 10^3$ CFU/mL, with a significant association found between proton pump inhibitor (PPI) use and SIBO. At $\geq 10^5$ CFU/mL, 33.3% tested positive for SIBO, and neither PPI use nor surgical history was associated. Breath testing revealed a 30.6% positivity rate for MO, predominantly characterized by intestinal methanogen overgrowth (IMO). There was no association between control of diabetes and positive MO tests.

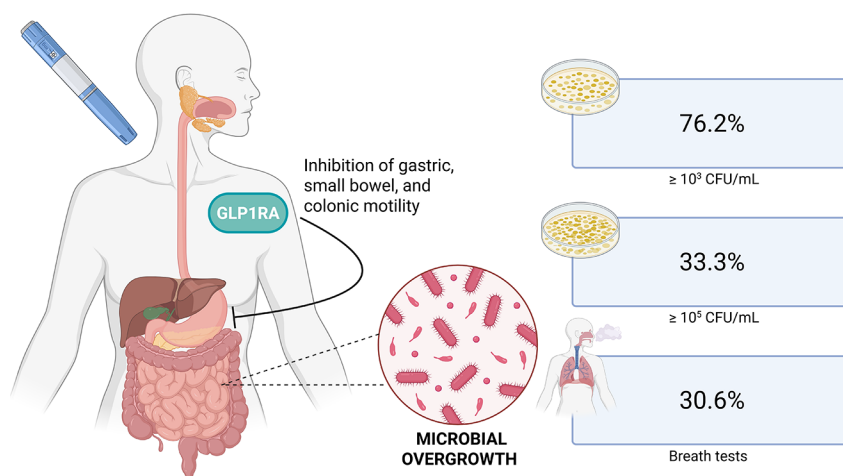
Conclusion: GLP1RA use is associated with SIBO, although diabetes may be a confounder.

Keywords

small intestinal bacterial overgrowth, SIBO, GLP-1 receptor agonists, microbiome, motility, obesity

Visual Abstract

GLP-1 receptor agonist use is associated with small intestinal bacterial overgrowth and intestinal methanogen overgrowth



Key Learning Points

1. SIBO is common among patients on GLP1RAs, with positive aspirate cultures occurring in 33.3% or 76.2%, depending on diagnostic threshold.
2. Breath tests are positive in 30.6%, 91.% of which reflect IMO.
3. While most of the patients in the cohort had diabetes, there was no difference in MO risk between those with adequately and poorly controlled diabetes.

Introduction

Glucagon-like peptide-1 (GLP1) receptor agonists (GLP1RAs) have emerged as first-line treatments for diabetes mellitus and obesity due to their considerable efficacy. Utilization of this class of medications is steadily increasing, and there are active investigations into their use for several other conditions ranging from cardiometabolic to neurologic. The multisystem effects of GLP1RAs derive from several mechanisms, including actions targeting the brain and others which exert a direct effect on the gastrointestinal tract. Alterations in gastrointestinal motility are especially important. GLP1RAs are well known to inhibit gastric motility. Preclinical models and human studies have demonstrated that GLP-1 and GLP1RAs delay gastric emptying¹ and increase accommodation.² Short-acting exenatide has been shown using gastric emptying scintigraphy to delay gastric emptying;³ similar effects have been seen with long-acting liraglutide.⁴ Upper gastrointestinal side effects such as nausea and vomiting are common, occurring in about 25% of patients.¹ Consequently use of these agents has been associated with gastroparesis, portending a 3.67 fold increased risk compared with bupropion-naltrexone.⁵

The inhibitory effect on motility is not limited to the stomach. GLP1RAs also delay small bowel transit, evidenced by a reduction in interdigestive migrating motor complexes and postprandial spike activity.¹ Exenatide significantly suppresses small intestinal transit.⁶

Less is known about GLP1RAs and colonic motility; however, preliminary evidence suggests an inhibitory effect, demonstrated by a reduction in circular contractions from full thickness muscular colon strips.⁷

Since GLP1RAs inhibit motility, it is possible that some adverse effects of these medications arise from gastrointestinal microbial overgrowth (MO). MO describes

symptoms attributed to overabundance of microbes in the small intestine. It is a secondary condition arising when physiologic defenses against overgrowth are disturbed. One such protective mechanism is gastrointestinal motility, preventing stasis and resultant overgrowth. MO has been associated with absence of the migrating motor complex^{8,9} and gastroparesis, with one meta-analysis showing the pooled prevalence of small intestinal bacterial overgrowth (SIBO) to be 41%.¹⁰ Constipation has also been associated with MO, specifically overgrowth of methanogenic archaea, termed intestinal methanogen overgrowth (IMO).¹¹ While it is not known whether IMO is the cause or effect of constipation, it has been associated with delayed small bowel and colonic transit.¹² It is plausible then that GLP1RAs, known to inhibit gastrointestinal motility, could predispose to MO. We sought to investigate the link between GLP1RA use and MO.

Methods

We queried the electronic health record to identify patients within Mayo Clinic prescribed GLP1RAs and then underwent testing for MO with either esophagogastroduodenoscopy with intestinal aspirate culture or breath test (glucose or lactulose). Cultures were considered positive if microbiota recovered was $\geq 10^3$ CFU/mL or $\geq 10^5$ CFU/mL. Small intestinal fungal overgrowth was diagnosed if yeast/fungi were $\geq 10^3$ CFU/mL. Breath tests were considered positive by a rise in hydrogen above baseline ≥ 20 ppm by 90 minutes or a methane level ≥ 10 ppm at any time point.¹³ Charts were also analyzed for diabetes, hemoglobin A1c (within 3 months of testing), use of a proton pump inhibitor (PPI), history of abdominal surgeries, and results of gastric emptying scintigraphy (GES). Adequate diabetes control was defined by a hemoglobin A1c of $<7\%$ within 3 months of MO testing.¹⁴ Descriptive statistics and regression analyses were performed in Microsoft Excel.

Results

We identified 99 patients prescribed GLP1RAs who underwent MO testing (Table 1).

Mean time from GLP1RA prescription to MO test was 538 days.

Sixty-three patients underwent testing via culture. Diabetes was present in 58. Twelve had undergone abdominal surgery (five Roux-en-Y gastric bypass, three

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Table 1. Demographic Information of Included Patients.

Demographic	Results
Age (years)	56.3 (range 27-90)
Sex	F 62 (62.6%) M 37 (37.4%)
Race/ethnicity	Caucasian 93 (94%) Black or African American 3 (3.2%) Pacific Islander 1 (1%) Unknown 2 (2%)
BMI (kg/m ²)	34.9
Hemoglobin A1c (%)	7 (range 4.2-12.5)
GLP-1 receptor agonist indication	Type II diabetes mellitus 91 (92%) Obesity 8 (8%)
EGD indication	Diarrhea: 22 Dyspepsia: 12 GERD: 9 Nausea and vomiting: 8 Abdominal pain: 4 Pre-bariatric surgery: 3 Bloating: 2 Abnormal imaging: 1 Vitamin and micronutrient deficiencies: 1 Dysphagia: 1

Note. Age is at time of MO test and BMI is at time of first GLP-1 receptor agonist prescription. Hemoglobin A1c is within 3 months of MO testing.

cholecystectomy, one sleeve gastrectomy, one hysterectomy, one partial colectomy, one hepatic wedge resection) and 29 were taking a PPI. GES had been performed in 12 patients.

Twenty-four patients were prescribed semaglutide, 23 dulaglutide, and 16 liraglutide. SIBO was positive in 48 patients (76.2%) at $\geq 10^3$ CFU/mL and 21 (33.3%) at $\geq 10^5$ CFU/mL. PPI use was associated with SIBO at $\geq 10^3$ CFU/mL ($P=0.04$), but not at $\geq 10^5$ CFU/mL ($P=0.162$). All 12 patients with surgery had SIBO at $\geq 10^3$ CFU/mL, so statistical analysis could not be performed; surgery was not associated with SIBO at $\geq 10^5$ CFU/mL ($P=0.525$). Twelve patients in the aspirate group had undergone GES. Four (33.3%) had delayed gastric emptying. SIBO was present in all four at $\geq 10^3$ CFU/mL, and only one at $\geq 10^5$ CFU/mL. There was no significant differences in SIBO results comparing controlled and inadequately controlled diabetes (Table 2).

Among cultures which grew $\geq 10^3$ CFU/mL, 26 grew both aerobic and anaerobic bacteria, 20 grew only aerobic, and two grew only anaerobic. Five had a specific organism identified: *Escherichia coli* in a patient with $\geq 10^5$ CFU/mL, *Klebsiella pneumoniae* complex ($\geq 10^5$ CFU/mL), *Escherichia coli* ($\geq 10^5$ CFU/mL), *Klebsiella pneumoniae* ($\geq 10^3$ CFU/mL), and *Klebsiella*

Table 2. Comparison of MO test results by status of diabetes control, where controlled diabetes is defined as a hemoglobin A1c of less than 7% within 3 months of the test.¹⁴

MO testing modality	Positive tests in controlled diabetes (number, %)	Positive tests in inadequately controlled diabetes (number, %)	Statistics
$\geq 10^5$ CFU/mL	8/34 (23.5%)	11/24 (45.8%)	$P=0.09$
$\geq 10^3$ CFU/mL	17/25 (68%)	21/24 (87.5%)	$P=0.17$
Breath tests	6/14 (42.9%)	5/19 (26.3%)	$P=0.546$

Note. P values reflect Fisher's exact test.

Oxytoca/Raoultella Ornithinolytica/Planticola in a patient with ($\geq 10^3$ CFU/mL).

Two (3.2%) grew ≥ 1000 CFU/mL of yeast, both in patients with SIBO ($\geq 10^5$ CFU/mL).

Thirty-six patients underwent breath testing. Of these, 33 had diabetes. Twelve had undergone surgery (five Roux-en-Y gastric bypass, four cholecystectomy, two partial colectomy, and one lap band) and 24 were on PPI. Fifteen were prescribed dulaglutide, 13 liraglutide, and eight semaglutide. Eleven (30.6%) had positive tests: nine (81.8%) due to elevation in methane, one (9.1%) due to hydrogen, and one (9.1%) due to both. There was no association between PPI use and positive breath test ($P=1.27$) or prior surgery and positive breath test ($P=0.967$). Seven had undergone GES; one (14.3%) was delayed and had IMO. There was no significant differences in breath test results comparing controlled and inadequately controlled diabetes (Table 2).

Two patients were identified who underwent MO testing after being prescribed the dual GLP-1/GIP agent tirzepatide. One had a negative breath test, the other grew 1000 to 10,000 CFU/mL.

Discussion

This is the first study to assess the relationship between GLP1RA use and MO. Of 99 patients prescribed GLP1RAs, 76.2% had SIBO at $\geq 10^3$ CFU/mL and 33.3% at $\geq 10^5$ CFU/mL, and 30.6% of patients had positive breath tests. Almost all positive breath tests reflected IMO. The pooled prevalence of MO across diagnostic modalities is 59.6% or 32.3% depending on aspirate threshold. Mean time from first prescription to SIBO test is about a year and a half. PPI use was associated with MO at $\geq 10^3$ CFU/mL, but not at $\geq 10^5$ CFU/mL.

The culture threshold to diagnose SIBO is debated. Historically, the gold standard has been $\geq 10^5$ CFU/mL; however, $\geq 10^3$ CFU/mL shows good correlation between the presence of pathogenic microbial signatures, metabolomic profile, and symptoms.¹⁵ In our study, there was concordance between SIBO results at the stricter $\geq 10^5$ CFU/mL

threshold (33.3%) and breath test positivity (30.6%), indicating that in this cohort, perhaps $\geq 10^3$ CFU/mL is overly sensitive and insufficiently specific. Additionally, PPIs were a confounding factor at $\geq 10^3$ CFU/mL but not at $\geq 10^5$ CFU/mL, which could support this higher threshold for diagnosis in this cohort. However, this conclusion is limited by the fact that breath testing among patients on GLP1RAs likely has a high false negative rate, due to insufficient substrate reaching the small bowel by the end of the test in the setting of slow gastric and small bowel transit.¹⁶

Culture data matches prior studies on SIBO, which have found predominance of Enterobacteriaceae and specific disruptor taxa such as *Klebsiella*, *Escherichia*, and *Enterococcus*.¹⁵

Diabetes is an important confounder, since 92% of patients had diabetes. Due to the small number of patients without diabetes, we were unable to control for this variable. Diabetes itself is associated with MO. A meta-analysis identified a pooled prevalence of 29%, with higher rates of positivity for aspirates (39%) than breath tests (29% for glucose, 31% for lactulose). The odds ratio for MO was 2.91 compared to patients without diabetes.¹⁷ A more recent study evaluating patients with diabetic gastroenteropathy found a SIBO prevalence of 45% diagnosed via culture ($\geq 10^5$ CFU/mL). SIBO was not associated with abnormal gastric emptying.¹⁸ In another study of patients with diabetic gastroenteropathy, symptoms in response to duodenal lipid infusion were not reduced (and in fact were slightly increased) with GLP1RA exendin.¹⁹ Our results at $\geq 10^5$ CFU/mL match those of the meta-analysis. Notably, the meta-analysis did not control for potential confounders such as medication use. Our prevalence is lower than that of the diabetic gastroenteropathy study, presumably because diabetic gastroenteropathy may portend a higher risk of SIBO than diabetes without related gastrointestinal dysfunction. The fact that our results align so closely with previous studies on diabetes and MO, it is possible that our findings are driven primarily by diabetes and not by GLP1RA use. While we could not adequately control for diabetes, it is notable that the mean hemoglobin A1c level around time of MO testing was 7%, which is the American Diabetes Association target for diabetes control.¹⁴ Furthermore, in our subanalysis of patients with diabetes who had recent hemoglobin A1c measurements, we found no statistically significant difference in MO among patients with controlled versus inadequately controlled diabetes, suggesting the effect may indeed be driven by the GLP1RA.

Obesity has also been hypothesized as a risk factor for MO. A meta-analysis found a twofold likelihood of MO among individuals with obesity; however this finding was not statistically significant. But when analyzing only data from Western countries, the risk was threefold and significant. Notably, this study did not assess for confounders such as PPI use.²⁰

PPI use was found to be a confounder at $\geq 10^3$ CFU/mL but not at $\geq 10^5$ CFU/mL, which may reflect that this threshold is more specific for SIBO, particularly in those with PPI use.

Our sample size of patients who had undergone GES was too small to evaluate whether delayed gastric emptying is associated with MO. A meta-analysis of six studies found a MO prevalence of 41% in gastroparesis. It is interesting however that since most of the transit studies in our cohort were normal, perhaps MO associated with GLP use arises from other mechanisms than just delayed gastric emptying, such as delayed small bowel and potentially even colonic transit. This may be supported by our breath test data. All but one of the abnormal breath tests showed IMO, which has been associated with delayed small bowel and colonic transit.¹²

There are several limitations to this study. The retrospective nature limits our ability to establish causality and may introduce biases related to incomplete data and reliance on historical records. For example, assessing medication use by prescription may not be accurate, as this does not reflect whether a patient was actually taking it (due to cost and insurance coverage, nonadherence, adverse effects, etc.) Furthermore, we were unable to directly assess our original question of the relationship between GLP1RA use and MO since almost all patients in the cohort also had diabetes, which may be a confounding factor. Due to the small sample size and the fact that some patients had been prescribed more than one GLP1RA, we were unable to perform subanalyses by specific agent. Finally, we do not have information regarding gastrointestinal transit times, which is important both mechanistically for the development of MO and contextually for the interpretation of breath tests.¹⁶ Future studies should take a prospective approach to assessing the risk of MO among patients without diabetes on GLP1RA, controlling for important potential confounders such as PPI use and gastrointestinal dysmotility.

Conclusion

Among patients taking GLP1RAs undergoing MO testing with either intestinal aspirate culture or breath test, most of whom had diabetes, the pooled prevalence of MO is 59.6% or 32.3% depending on an aspirate threshold of $\geq 10^3$ CFU/mL or $\geq 10^5$ CFU/mL, respectively.

Abbreviations

GLP1: Glucagon-like peptide-1
 GLP1RA: Glucagon-like peptide-1 receptor agonist
 MO: Microbial overgrowth
 SIBO: Small intestinal bacterial overgrowth
 IMO: Intestinal methanogen overgrowth

Author Contributions

J.A.D. conceived of manuscript, obtained microbial overgrowth data, designed methodology, performed data analysis, and wrote and edited manuscript. T.F. obtained GLP-1 receptor agonist data, designed methodology, and edited manuscript. M.J. obtained microbial overgrowth data and edited manuscript. L.O.V. obtained microbial overgrowth data and edited manuscript. X.J.W. designed methodology, supervised analysis, and edited manuscript.

Data Availability

Available upon request.

Declaration of Conflicting Interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: Dr. Damianos has received clinical trial research funding from ExeGi Pharma as well as consulting fees, and speakers bureau fees from i-Health. All other authors declare no conflicts of interest.

Ethical Approval

This study was approved by the Mayo Clinic Institutional Review Board.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Use of Artificial Intelligence

No artificial intelligence (AI) was used in the preparation of this manuscript.

Patient Consent

There was no patient contact involved in this study.

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